Nuno Alexandre Goncalves

Software Engineer | Python, Java & Data Systems | Commodities Trading Systems
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SUMMARY

Software Engineer with 4 years **building production systems** inside a large-scale commodities trading environment for traders and vessel managers (BP, via Infosys).
Built backend services for a commodities arbitrage platform used by front-office trading desks, shipped a natural-language operations tool with an **estimated $2M** in cost savings, and **cut query latency by 85%** on live trading data. Interested in technically demanding roles building systems at scale.

SKILLS

Languages: Python, SQL, Java, JavaScript
Infrastructure: Podman, AWS EC2, AWS S3, Azure, Linux, Git
Backend & Web: Spring Boot, REST APIs, React
AI/ML: RAG, OpenAI, Azure AI Search, LangChain, Databricks

WORK EXPERIENCE

Software Engineer, Infosys (Client: BP) Aug 2022 – Present
London, UK

- Built backend APIs **(Java, Spring Boot)** and React UI components integrating reference data into a commodities arbitrage platform used by front-office trading desks.
- Engineered a vector search layer **(Azure AI Search)** over live vessel sensor data, cutting query latency by 85% and eliminating full table scans across downstream pipelines.
- Designed and shipped an LLM-powered control panel **(Python, Azure OpenAI)** that allowed vessel managers to query live shipment data in natural language, generating **~$2M in estimated cost savings**.
- Built a deterministic LLM chatbot **(LangChain, Databricks)** over BP's master reference system, resolving ambiguous queries via follow-up questions without costly direct **Databricks** calls.
- Built a RAG platform **(Azure AI Search, Azure OpenAI)** indexing 2M+ enterprise documents, emails, and scanned images via a multimodal ingestion pipeline, enabling natural-language retrieval across BP's operational documents.
- Awarded the **Rising Star**, **TechZooka**, **Rise Insta Award** (Technical Excellence & Mentorship)

Operations Intern, Echobox Jul 2021 – Sept 2021
London, UK

- Automated business development workflow processes (**Java, Salesforce Apex)** for email outreach, reducing overall lead handling time by **30%** for BDRs.
- Audited 2,000+ customer accounts, validating terms and conditions and managing Salesforce data to support quarterly onboarding, improving Echobox's customer insights.

LifeSkills Youth Advisory Council Member, Barclays Jan 2020 – May 2023
London, UK

- Advised senior stakeholders (including Karren Brady) on UX and curriculum strategy for a platform with 1M+ annual users.

EDUCATION

University of Bath Sept 2019 – July 2022
BSc Computer Science – First Class Honours
Modules: Advanced Algorithms & Complexity, AI, Machine Learning, Data Structures & Algorithms, Compilers

PROJECTS

Turing Machine Lambda Calculus Interpreter (JavaScript, Bootstrap)

- Built a live web interpreter for bidirectional translation between lambda calculus and Turing machines, based on Dal Lago & Martini's formal methods, with Jest test coverage.